Project for Bayesian Inference

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Model

Likelihood

$$Y_i \mid \theta, \sigma^2 \overset{iid}{\sim} \mathcal{N}(\theta, \sigma^2), \quad i = 1, \ldots, n$$

Case 1 (baseline prior)

$$\theta \sim \mathcal{N}(\mu_0, \tau_0^2), \qquad \tilde{\sigma}^2 = \frac{1}{\sigma^2} \sim \text{Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0}{2}\sigma_0^2\right)$$

Case 2 (hierarchical prior)

$$\theta \sim \mathcal{N}(\mu_0, \tau_0^2), \quad \tilde{\sigma}^2 \mid \sigma_0^2 \sim \text{Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0}{2}\sigma_0^2\right)$$

$$\sigma_0^2 \mid \beta \sim \text{Gamma}(a, \beta), \qquad \beta \sim \text{Gamma}(b_1, b_2)$$

- Case 1 fixes σ_0^2 using prior information.
- Case 2 treats σ_0^2 as a random parameter with an additional hierarchical layer.
- This allows uncertainty in the variance prior to propagate to posterior inference.

Prior sensitivity (factorial design)

$$a \in \{0.1, 2, 10\}, \qquad (b_1, b_2) \in \{(0.1, 10), (2, 200), (10, 1000)\}$$

All combinations yield $3 \times 3 = 9$ prior settings to assess prior robustness on $(\theta, \sigma^2, \sigma_0^2, \beta)$.

Gibbs Sampler: Case 1 & Case 2 (full conditionals)

Likelihood $Y_i|\theta,\sigma^2 \sim \mathcal{N}(\theta,\sigma^2)$, $\tilde{\sigma}^2 = 1/\sigma^2$, $\bar{y} = \frac{1}{n}\sum_i y_i$, $\mathrm{SSE}(\theta) = \sum_i (y_i-\theta)^2$.
Case 1 priors $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$, $\tilde{\sigma}^2 \sim \mathrm{Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0}{2}\sigma_0^2\right)$.

- $\theta|\tilde{\sigma}^2, y \sim \mathcal{N}\left(\frac{n\bar{y}\tilde{\sigma}^2+\mu_0/\tau_0^2}{n\tilde{\sigma}^2+1/\tau_0^2}, \frac{1}{n\tilde{\sigma}^2+1/\tau_0^2}\right)$
- $\tilde{\sigma}^2|\theta, y \sim \mathrm{Gamma}\left(\frac{n+\nu_0}{2}, \frac{\mathrm{SSE}(\theta)}{2} + \frac{\nu_0\sigma_0^2}{2}\right)$

Case 2 priors $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$, $\tilde{\sigma}^2|\sigma_0^2 \sim \mathrm{Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0}{2}\sigma_0^2\right)$, $\sigma_0^2|\beta \sim \mathrm{Gamma}(a,\beta)$, $\beta \sim \mathrm{Gamma}(b_1, b_2)$.

- $\theta|\tilde{\sigma}^2, y \sim \mathcal{N}\left(\frac{n\bar{y}\tilde{\sigma}^2+\mu_0/\tau_0^2}{n\tilde{\sigma}^2+1/\tau_0^2}, \frac{1}{n\tilde{\sigma}^2+1/\tau_0^2}\right)$
- $\tilde{\sigma}^2|\theta, \sigma_0^2, y \sim \mathrm{Gamma}\left(\frac{n+\nu_0}{2}, \frac{\mathrm{SSE}(\theta)}{2} + \frac{\nu_0\sigma_0^2}{2}\right)$
- $\sigma_0^2|\tilde{\sigma}^2, \beta \sim \mathrm{Gamma}\left(a + \frac{\nu_0}{2}, \beta + \frac{\nu_0}{2}\tilde{\sigma}^2\right)$
- $\beta|\sigma_0^2 \sim \mathrm{Gamma}\left(a + b_1, b_2 + \sigma_0^2\right)$

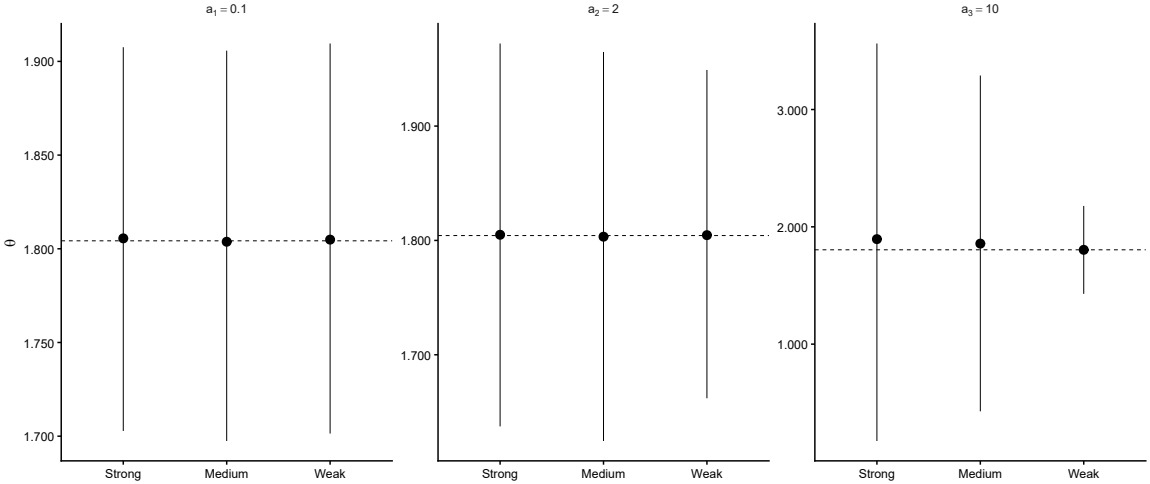
Iteration

Initialize $(\theta^{(0)}, \tilde{\sigma}^{2(0)})$;
Case2 also $(\sigma_0^{2(0)}, \beta^{(0)})$.
For $s = 1, \ldots, S$
Case 1
$\theta^{(s)} \sim p(\theta|\tilde{\sigma}^{2(s-1)}, y)$
$\tilde{\sigma}^{2(s)} \sim p(\tilde{\sigma}^2|\theta^{(s)}, y)$
Case 2
$\theta^{(s)} \sim p(\theta|\tilde{\sigma}^{2(s-1)}, y)$
$\tilde{\sigma}^{2(s)} \sim p(\tilde{\sigma}^2|\theta^{(s)}, \sigma_0^{2(s-1)}, y)$
$\sigma_0^{2(s)} \sim p(\sigma_0^2|\tilde{\sigma}^{2(s)}, \beta^{(s-1)})$
$\beta^{(s)} \sim p(\beta|\sigma_0^{2(s)})$
Cycle: $\theta \to \tilde{\sigma}^2 \to \sigma_0^2 \to \beta$

Posterior Interval Inference

Posterior Inference for θ under Different Hyperpriors

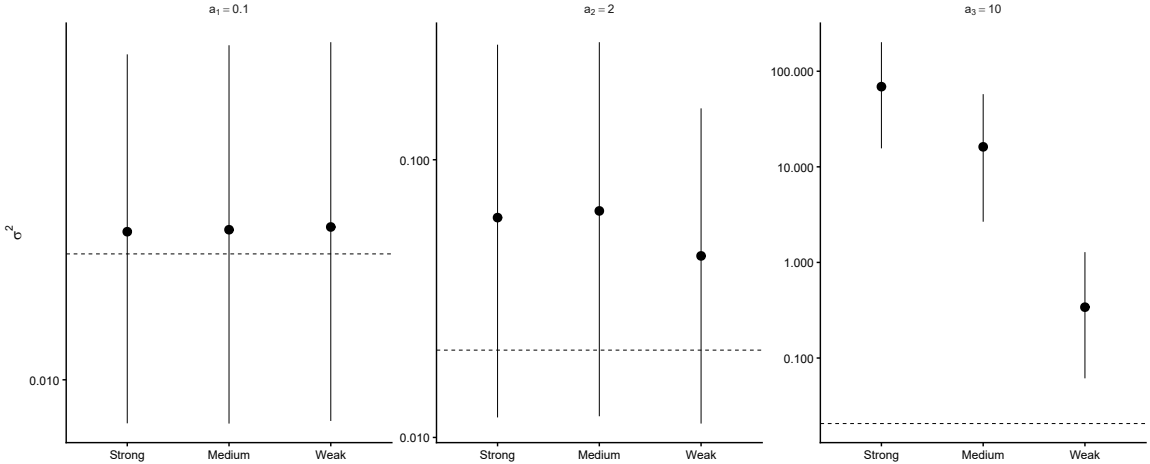
Posterior of θ: mean and 95% CI (dashed = Case 1 mean)



Case 2 hyperpriors: $\sigma_0^2 \sim \Gamma(a_k, \beta)$, $k = 1, 2, 3$, with $(a_1, a_2, a_3) = (0.1, 2, 10)$. $\beta \sim \Gamma(a_\beta, b_\beta)$ and $E[\beta] = 0.01$. Weak/Medium/Strong: $(a_\beta, b_\beta) = (0.1, 10), (2, 200), (10, 1000)$.

Posterior Inference for σ^2 under Different Hyperpriors

Posterior of σ^2: mean and 95% CI (log scale; dashed = Case 1 mean)



Case 2 hyperpriors: $\sigma_0^2 \sim \Gamma(a_k, \beta)$, $k = 1, 2, 3$, with $(a_1, a_2, a_3) = (0.1, 2, 10)$. $\beta \sim \Gamma(a_\beta, b_\beta)$ and $E[\beta] = 0.01$. Weak/Medium/Strong: $(a_\beta, b_\beta) = (0.1, 10), (2, 200), (10, 1000)$.

Mixing

Inference for mixing

MCMC diagnostics (Case2): green = better mixing

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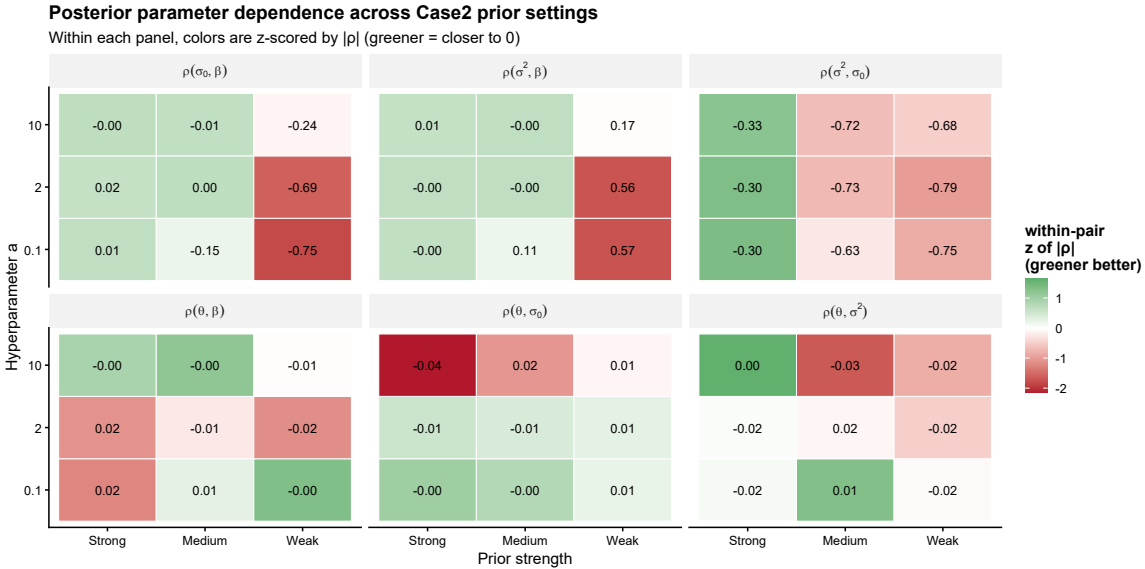
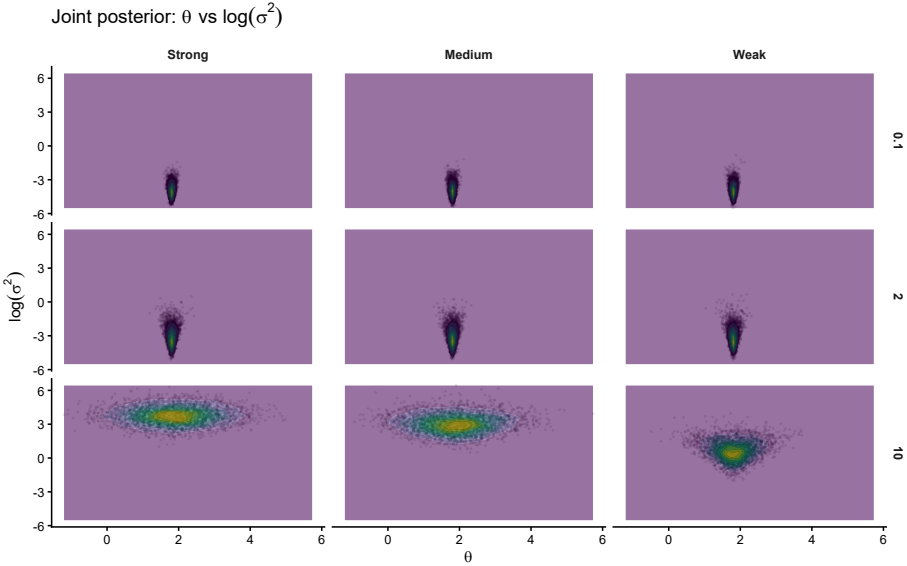
Joint Posterior inference

Correlation

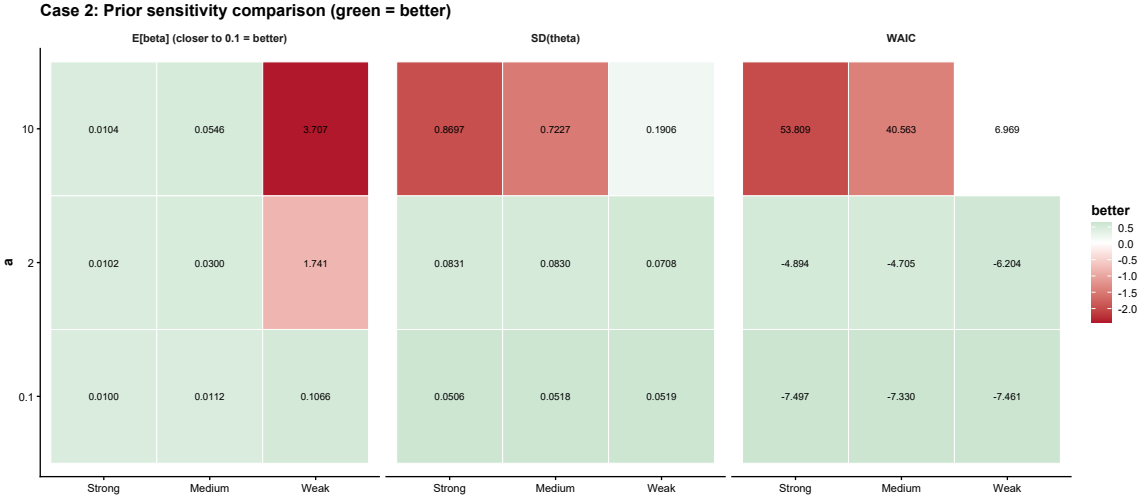


Joint density

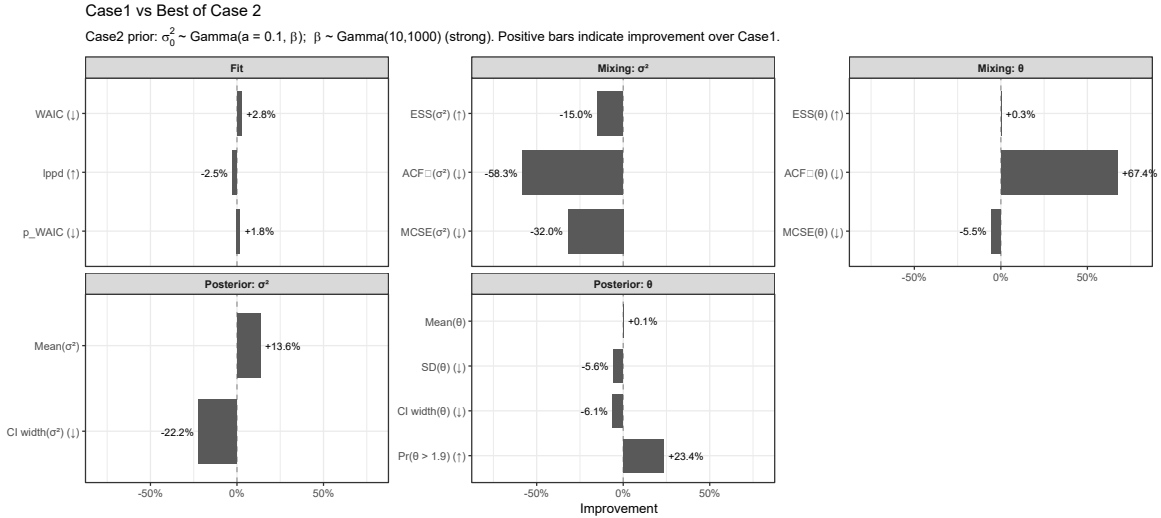


Model Comparison

Model comparison among Case 2



Comparing Case 1 to the best model in Case 2



Takeaway

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- When a is small (e.g. $a = 0.1$), the Gamma prior becomes **highly heavy-tailed** and places substantial probability on both very small and very large variance values.
- This can make the variance hierarchy unstable and increase posterior dependence among variance parameters.
- A **stronger prior on** β helps regularize the model by restricting the scale of σ_0^2 and stabilizing posterior inference.

Implication

- Small $a \Rightarrow$ diffuse variance prior.
- Stronger β prior $\Rightarrow$ stabilizes the hierarchical variance structure.

- Case 2 provides slightly improved model fit and more stable inference for θ.
- Some mixing diagnostics for σ^2 deteriorate slightly, indicating a trade-off between hierarchical flexibility and sampling efficiency.